



SafeStep

Project Progress & Decision Log

Campus safety companion app — Nelson Mandela University, Summerstrand

Flutter + Supabase (Postgres / Auth / RLS / Realtime / Storage)

Compiled 2026-09-16

Source: full build session history and git commit log (repo: SafeStep)

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1. Project Overview

SafeStep is a hackathon-prototype campus safety companion app for Nelson Mandela University's four Summerstrand campuses (South, North, 2nd Avenue, Ocean Sciences). It treats campus grounds and the surrounding off-campus suburb as one continuous safety area, with two responder organisations (NMU Campus Security and a private security company) routed by zone rather than by campus boundary.

One Flutter codebase serves three surfaces: a mobile app shared by students and responders (role resolved by which database table the signed-in user's row lives in, not by a separate build), and a web-only admin dashboard (same codebase, web target). The backend is real Supabase — real Auth, real Postgres with Row Level Security, real Storage — seeded throughout with fictional/simulated people, alerts and reports rather than mocked infrastructure.

This log reconstructs, in order, what was decided, what was built, what broke, and how each fix was verified — drawn from the project's git history (with real commit timestamps) and the build session transcripts.

2. Day 1 — 2026-09-13 — Initial Build

2026-09-13, 00:13

Initial checkpoint: student + responder apps NEW

- Full student app: registration wizard, SOS / silent alert, Walk With Me, incident reporting, trusted contacts, walking groups (regulated preset-message chat), safety resources, profile/settings, account deletion.

- Full responder app: alert feed, alert detail with status actions, history, profile.
- Role-based routing via database table membership (`students` / `responders` / `admins`), with an admin-invite + self-activation flow for responders and admins.
- RLS policies and SQL migrations 0001–0008 committed alongside the app for documentation, aligned against `docs/scope.md` and `docs/hobjectives.doc`.

commit 9990577

2026-09-13, 22:21

Admin web dashboard, SOS reliability, real map + Walk With Me routing NEW

FIXED

- Built the Admin dashboard (Flutter Web): overview, live alerts, responder/admin management, zones, safety broadcasts, resources, incident reports, walking-group approval, emergency contacts, and a student directory.
- **Fixed:** a Timer/AnimationController race in the SOS hold button that could silently drop activation after a full 3-second hold; added a 3-alerts-per-10-minutes rate limit with a 1-hour ban (device-local at this point — later reworked, see 2026-09-16).
- **Fixed:** a silent RLS gap where a student's SOS alert saved correctly but never reached any responder's feed — students had no read access to the `responders` table needed for fan-out.
- Replaced the zone-list Map tab with a real interactive OpenStreetMap view (`flutter_map`) with risk-coloured zone markers.
- Built the Walk With Me companion-accept flow end-to-end: invite → real-time-poll acceptance → a real OSRM-routed path between start location and destination.
- **Fixed:** the same `setState(() => future = ...)` bug pattern (a `setState` callback returning a `Future`) swept across roughly 12 screens.
- **Fixed:** the INTERNET permission was only declared in the debug manifest — a release build would have had no network access at all.
- Populated demo-data gaps: trusted contacts with phone numbers for every student, medical/vehicle/mobility info for a realistic subset, responder phone numbers, zone coordinates.

commit 1ab354d

2026-09-13, 22:44

Offline SOS fallback, real map tiles, location button NEW FIXED

- SOS now checks real connectivity before attempting to create an alert: with no data connection it routes straight to Call Security (with a clear banner) instead of holding for 3 seconds toward a doomed request. Call Security caches emergency numbers locally so it still works with zero connection.
- **Why this took two attempts:** two tile providers (Google, then CartoDB) turned out to require an API key/billing for real app use despite looking free in an isolated curl test. Settled on Esri's

World_Street_Map service, verified as a genuine ~2.5 KB JPEG tile image before committing to it, not just an error page with the right HTTP status.

- Added a "my location" button on the Map tab, centring on the student's real GPS position with a live marker.
- Seeded 12 more zones across Summerstrand with a real spread of risk levels — previously zero zones were marked "high", so there was no red-zone demo case at all.

commit ba399c2

3. Day 2 — 2026-09-16 — Feature Backlog & Refinement

The user opened this session with a single large backlog message covering roughly a dozen distinct requests: alert timestamps, a new "Safe Ride" vehicle safety-check feature, the SOS ban wrongly blocking every student instead of one, a post-signup login glitch, zone-scoped responder visibility, an NMU brand touch, a login-email format change, a Walk With Me rename, fuller privacy-notice content, group member names, message-list ordering, and an evaluation of voice-activated SOS. Each is tracked below in the order it was actually resolved.

2026-09-16, 13:10

Signup wizard race, per-student SOS ban, UTC timestamp bug, zone-scoped alerts

FIXED

- **Fixed — signup wizard glitch** ("after creating an account I had to hit continue then back to reach step 2"): `animateToPage()` was called in the same tick as `setState()`, before the `PageView` was necessarily attached on the very first step transition. Deferred to a post-frame callback.
- **Fixed — SOS ban blocking everyone** ("if a student misuses the SOS button, temporarily block that one student, currently it blocks everyone"): reworked from device-local `SharedPreferences` to a server-side ban keyed to `student_id`, computed off the student's own real `alerts` history. Verified end-to-end with two separate accounts on the same device, proving no cross-contamination.
- **Fixed — the "1 minute reads back as longer" bug**: root-caused to a UTC/local timestamp mismatch. The database session timezone is UTC, and Postgres interprets any offset-less ISO-8601 string as already being UTC — but `DateTime.now().toIso8601String()` in Dart produces a *local*, offset-less string. Every such write was silently stored skewed by the device's own UTC offset. Fixed at all 10 call sites with `.toUtc()`, confirmed against a live database session (`current_setting('TimeZone') → UTC`).
- **New — zone-based alert routing** (scope.md §3): alerts now carry a real `zone_id` via nearest-point lookup against real zone coordinates; responder fan-out is scoped to that zone's assigned responders (failing open to all active responders if no zone match exists). Verified a South Campus responder sees a South Campus alert while a North Campus responder does not.
- Added formatted local timestamps to every alert view on both the responder and admin sides.

commit 268a948

2026-09-16, 13:19

Safe Ride feature, "Walk With Me" → "Safe Walks" rename NEW

- **New — Safe Ride:** a student checks a lift/taxi's number plate against real on-file offense records before getting in. Risk is classified by *both* the magnitude and count of offenses — any single severe offense, or 3+ moderate ones, is "danger" (red); 1–2 moderate or 3+ minor is "moderate" (orange); otherwise "clean" (green) — not just a raw offense count. Includes an immediate "report this car" flow.
- Seeded 20 vehicle records with realistic South African driver/vehicle details and a genuine offense-profile spread: 8 clean, 7 moderate, 5 danger.
- **Verified, not assumed:** *confirmed a brand-new report takes effect immediately (no moderation queue), and — critically — confirmed RLS actually blocks a student from inserting a report with `source='seed'` (a real 403), proving the spoof-protection genuinely works rather than just looking correct in the UI.*
- Swapped the Home screen's "Map" quick-action tile for "Safe Ride" — Map was redundant there (already a bottom-nav tab one tap away), and scope.md caps Home at exactly 4 tiles with no scrolling.
- Renamed "Walk With Me" → "Safe Walks" across every user-facing string (screen titles, notification channel description, companion-invite copy, preferences label), deliberately leaving internal identifiers (notification channel ID, SharedPreferences key) unchanged.

commit 6517f2e

2026-09-16, 13:24

Group member names RLS gap, message auto-scroll FIXED

- **Fixed:** group members could only ever see their *own* name in the member list and chat — the `students` SELECT policy didn't cover fellow group members, so the embedded name join silently returned `null` for everyone else. This is the same class of bug as an earlier companion-invite name gap. Verified 1 of 21 members showed a real name before the fix; added a narrow RLS policy scoped to shared group membership; verified 21 of 21 after.
- **Fixed — "recent messages should be at the bottom":** the sort order was already correct (oldest → newest), the real gap was that the screen never auto-scrolled there, leaving the newest message off-screen. Added auto-scroll-to-bottom on load and after sending.

commit 1b93e23

2026-09-16, 13:28

Full privacy-notice content, subtle NMU theme touch NEW

- Rewrote the privacy notice to explain the actual three-tier visibility model (restricted/emergency-only, aggregate-only, operational) in plain language with concrete examples, plus dedicated sections for trusted contacts, anonymous reports, walking groups, Safe Ride report attribution, and data retention/deletion rights.
- Shifted the primary theme colour to a navy approximating NMU's actual brand palette (navy + gold — NMU publishes only Pantone codes, not hex, so this is a close approximation) and added a single gold accent line on the Welcome screen, per the request for a "*short, polite, lowkey*" brand touch. The safety-critical alert/caution/safe colours were deliberately left untouched.

commit e08c6d0

2026-09-16, 13:41

Theme toggle, SOS accessibility/haptics, visible support button NEW

This batch and the next were built against a usability checklist proposed mid-session and explicitly approved by the user — *"yeah have an option to choose, system should be default, add all these except for the larger touch targets"* — i.e. every item below except larger touch targets.

- Added an explicit System/Light/Dark override in Settings, defaulting to System and persisted via SharedPreferences (previously the app only ever followed the OS setting with no override).
- Added screen-reader semantics to the SOS button (label + hint) and its countdown sheet (live-announcing seconds remaining), and haptic feedback on SOS hold-start, SOS activation, incident-report submission, and every responder alert status action.
- Added a visible "Get Support" entry point (header icon, warmer tone than Call Security's urgent framing) — reuses the existing `emergency_contacts` data.

commit a64b197

2026-09-16, 13:50

Skeleton loaders, pull-to-refresh, empty states, undo-on-delete NEW

- New reusable skeleton-loading widgets (pulsing-opacity placeholders, no external shimmer dependency), replacing bare spinners on initial list-loading across trusted contacts, groups, resources, my reports, alerts, companion invites, and responder history.
- Added pull-to-refresh to every list screen missing it.
- Replaced plain "no X yet" empty states with an icon, heading, and concrete next-step guidance across the same set of screens.
- Removing a trusted contact or leaving a group now takes effect immediately with an Undo action on the confirming snackbar, instead of a blocking confirm dialog in front of an irreversible delete.

commit 7ea52c6

2026-09-16, 13:56

Biometric app-lock (Face/Touch ID / PIN fallback) NEW

- Locks the app on launch and whenever it returns from the background, scoped to native platforms only (never the web-only admin surface). Off by default; a Settings toggle turns it on, gated behind a successful biometric/PIN check before it can be enabled at all.
- Android requires `FlutterFragmentActivity` rather than `FlutterActivity` for the biometric prompt to host correctly, plus the `USE_BIOMETRIC` manifest permission.

commit 98e85bf

2026-09-16, afternoon

Student/staff login-email migration DB CHANGE — not a code commit

- Migrated real Supabase Auth login emails to the requested institutional format: students → `s<9 random digits>@mandela.ac.za`, responders → `s<5 random digits>@mandela.ac.za`, admin left unchanged.
- 90 students + 10 responders migrated (100 accounts), zero write failures. Both `auth.users.email` and the mirrored `students / responders` table row were updated together via the Supabase service-role Admin API, so they can never drift apart. The 4 responders still in `invited` status (no auth account yet) had only their table row updated, so future self-activation will match on the new address.
- **Verified, not assumed:** sampled 5 students + 5 active responders — logged in with the new email and each account's real seed password (real `200`), and confirmed the old email now rejects login (real `400`), 10 of 10.
- No app code changes were required — the login form only ever validated for the presence of "@", no domain restriction existed.

2026-09-16, afternoon

Feature-gap audit; voice-activated SOS evaluated and explicitly skipped

- Audited the live code against `docs/scope.md` (which had drifted out of date in places) rather than trusting the document at face value. Confirmed genuinely built and working: photo upload on incident reports (wired to real storage), walk-session missed-check-in escalation, silent alert mode, QR scan screen, offline connectivity fallback, medical info card.
- Confirmed one real functional gap: acknowledging an alert flips the shared `alerts.status` globally, so other responders do see the status change, but the UI never surfaces *who* acknowledged it (`scope.md` §3/§6's "Acknowledged by [name/org]" is not yet built) — see §7 Open Items.
- Also flagged: notifications rely on `flutter_local_notifications` only (no FCM/OneSignal), so delivery only happens while the app is open/foregrounded on a device — fine for a live demo, not true push.
- **Voice-activated SOS — evaluated, not built, by explicit user choice.** True passive background listening was assessed as impractical for this prototype (OS-level background mic

throttling, battery cost, and a real privacy concern for an always-listening safety app). A scoped-down alternative — an explicitly-activated, time-bounded listening session using the free `speech_to_text` package — was proposed as buildable. The user chose to skip voice SOS entirely rather than build even the scoped version.

2026-09-16, 17:11

SafeStep rebrand: custom icon, branded auth screens, biometric escape hatch NEW

FIXED

- Designed and generated a navy/gold shield app icon, wired through `flutter_launcher_icons` for Android/iOS/macOS/Windows, and set the app display name to "SafeStep" everywhere (Android label, iOS bundle name, macOS/Windows product name and window title). Verified on-device via `flutter devices` and a live `flutter run` against the connected phone.
- The user then supplied their own logo (`assets/icon/safestep.jpg` , a blue shield with a white "S") to replace the generated one. Regenerated all platform icons from it, and — critically — deleted the now-stale Android *adaptive* icon files (a separate navy/gold layered icon) that would otherwise have silently overridden the new flat icon on Android 8+.
- Added the logo + "Your Safety Companion" tagline to the Welcome, Log In, and Create Account screens via one shared `AuthBrandingHeader` widget.
- Removed the now-duplicate "Safe Ride" entry from the drawer, since it already has a Home-screen tile.
- Expanded Safety Resources with 3 new guides (drink-spiking awareness, dorm/residence fire safety, bystander intervention) and added real, web-verified YouTube videos to 2 existing guides that were missing one (safe walking at night, supporting a friend) — all 10 published resources now have a linked video. DB change
- **Real incident — biometric lockout:** the user enabled biometric lock, the prompt failed on their device, and they were completely stuck behind the lock screen with no way back in — a serious gap for an app whose entire purpose is emergency access. Fixed two ways: (1) immediately unblocked the user by clearing the app's local data via `adb shell pm clear` , which also signs out every profile and resets the lock preference; (2) added a permanent "Sign out and turn off app lock" escape hatch directly on the lock screen so this can never happen again.

commit 6271ab6

2026-09-16, 17:11

Home-screen SOS widget; SOS tap-vs-hold race fixed NEW FIXED

- **New — SOS home-screen widget:** a native Android `AppWidgetProvider` (built on the `home_widget` plugin) placeable on the phone's home screen. Tapping it launches the app straight into a new Quick SOS screen — skipping Home entirely — but still requires the same full 3-second deliberate hold as the in-app button. The widget only removes navigation friction, not the accidental-activation safeguard.

- **Design note:** a fully "silent" background-fired SOS (no on-screen confirmation at all) was considered and rejected — it would violate the project's own established principle that panic activation must be deliberate, not one-tap, and risks a false alarm firing from an in-pocket tap with zero visible feedback.
- Extracted the shared "check connectivity → route to Call Security or Active SOS" logic out of the Home tab into one `handleSosActivated` helper, used by both the Home tab and the new widget entry point, so the two never drift.
- **Fixed — a real bug surfaced by testing:** a quick tap on the SOS button could start an unattended countdown on its own. Root cause: the SOS-ban gate check is a real network round-trip, awaited *before* the hold animation/timer started. If the finger lifted during that window, the release handler found nothing to cancel yet (no timer, no animation running) — and the hold then started anyway once the network check resolved, moments after the finger was already off the screen. Fixed with a `_pointerDown` flag re-checked immediately after the await, so a release during the gate check now correctly prevents the hold from ever starting.
- **Fixed — widget not appearing in the Android widget picker:** the provider was correctly registered at the OS level (confirmed via `adb shell dumpsys appwidget`), but Samsung's One UI launcher had cached a stale, empty widget list for the app across several debug reinstalls. Force-stopping the launcher process (`am force-stop com.sec.android.app.launcher`) made it re-scan and pick the widget up — no app reinstall or reboot needed.

commit 1c3da2e

4. Key Technical Decisions & Why

Decision	Why
Zone-based alert routing, not campus-based	scope.md's core design idea: two responder organisations (NMU Campus Security + a private security company) share one continuous map. Routing by zone lets both orgs' responders be notified together regardless of who covers what, and models real multi-unit dispatch ("Acknowledged by X" rather than duplicate untouched alerts).
Server-side, per-student SOS misuse ban	A device-local ban blocks every account tested on that device, not just the misusing student — wrong on a shared/test device and wrong in production, where one student's misuse should never lock out their roommate.
All alert-adjacent timestamps written as UTC	The Supabase/Postgres session timezone is UTC; an offset-less local ISO-8601 string is silently interpreted as already being UTC, skewing every stored time by the device's own offset. Explicit <code>.toUtc()</code> at every write site is the only reliable fix.

Esri World_Street_Map tiles over Google/CartoDB	Both alternatives looked free in an isolated request but required a billing-linked API key for real in-app use — caught by checking the actual image bytes returned, not just the HTTP status.
Safe Ride risk = magnitude <i>and</i> count of offenses	A raw offense count alone can't distinguish one severe incident from several trivial ones — the classification explicitly combines both so a single severe offense is never under-weighted.
Biometric lock never touches biometric data	The app calls the OS's own biometric prompt via <code>local_auth</code> and receives only a boolean result — no fingerprint/face data is ever accessible to, or stored by, the app. Confirmed to the user directly when asked.
SOS home-screen widget still requires a full hold	Removing navigation friction is valuable; removing the deliberate-activation safeguard is not — a home-screen tap is exactly the kind of accidental (in-pocket) trigger that safeguard exists to prevent.
Voice-activated SOS: evaluated, then skipped	True background listening was judged impractical (battery, OS throttling, privacy) for this prototype. A scoped, explicitly- activated version was offered but the user chose not to build even that, to keep the app's mic usage strictly opt-in per action.

5. Bugs Found & Fixed

Bug	Root Cause	How Verified
Post-signup wizard needed "continue then back" to reach step 2	<code>animateToPage()</code> called same-tick as <code>setState()</code> , before <code>PageView</code> attached	Deferred to <code>addPostFrameCallback</code> ; manual signup-flow retest
SOS misuse ban blocked every student on a device, not just the offender	Ban state stored in device-local <code>SharedPreferences</code> , not scoped per account	Rebuilt server-side; tested two accounts on one device for cross-contamination
Walk With Me timers read back far longer than the real elapsed time	Offset-less local timestamps silently stored as UTC by Postgres	Direct SQL check of session timezone + timestamp round-trip; fixed at all 10 sites
SOS alerts never reached any responder's feed	Students had no RLS read access to the <code>responders</code> table	Live end-to-end alert creation + responder-side feed check

Group members saw only their own name in the member list/chat	RLS SELECT policy on <code>students</code> didn't cover fellow group members, so the embedded join returned null	Real JWT test: 1/21 names visible before fix, 21/21 after
Newest group message appeared off-screen	Sort order was correct; there was simply no auto-scroll to the bottom	Manual retest of load + send
A quick tap on the SOS button could start an unattended countdown	The network-bound ban-gate check was awaited before the hold animation/timer started, so a release mid-check found nothing to cancel, and the hold began anyway once the check resolved	Root-caused from the reported symptom; fixed with a <code>_pointerDown</code> guard; <code>flutter analyze</code> clean
SOS home-screen widget didn't appear in the widget picker	Widget provider was correctly registered at the OS level, but Samsung's launcher had a stale cached widget list after repeated debug reinstalls	Confirmed registration via <code>adb dumpsys appwidget</code> ; fixed by force-stopping the launcher process
Biometric lock could permanently strand a user with no way back in	The lock screen had no fallback action if the biometric prompt genuinely failed	Reproduced live by the user; unblocked via <code>adb pm clear</code> ; added a permanent sign-out escape hatch

6. Feature Checklist (against docs/scope.md)

Student

- Registration wizard, trusted contacts (email/phone auto-link), SOS (hold-to-confirm) + silent alert + **home-screen SOS widget**
- Offline SOS fallback (native call + cached numbers/tips)
- Safe Walks: companion invite + accept, self-monitored with missed-check-in escalation, mid-journey extend
- Incident reporting (2-screen flow, anonymous option, photo upload to real storage)
- Continuous campus + Summerstrand risk map, walking groups (admin-approved), Safe Ride vehicle check, safety resources (10 guides, each with a real video), shuttle/library hours, QR emergency points, medical info card
- Settings: theme (System/Light/Dark), biometric app-lock, notification preferences

Responder

- Self-activation via admin invite; zone-scoped alert feed with student name/type/location/elapsed time/status; alert detail with map, medical-info banner, and Acknowledge/Dispatch/Resolve/False-Alarm actions; write report on resolve; history

Admin (web)

- Overview dashboard, responder/admin management, zone management, scheduled/retractable safety broadcasts, resource moderation, incident-report review, walking-group approval, emergency-contact management, student directory

7. Known Open Items

Item	Status
"Acknowledged by [name/org]" multi-responder awareness	Status flips correctly for every responder covering a zone; the UI does not yet show <i>who</i> acknowledged it. Confirmed via code read, not yet built.
Real push notifications (FCM/OneSignal)	Currently local-notifications only — delivery requires the app to be open/foregrounded on the recipient's device.
Full accessibility audit	SOS-specific semantics/haptics done; a full pass over every other screen has not been done.
Voice-activated SOS	Evaluated and explicitly skipped by the user — not built.
Scheduled safety broadcasts don't auto-fire at their scheduled time	Would need a server-side cron/edge function; already documented in-code as an accepted prototype limitation.

8. Commit Reference

Hash	Date/Time	Subject
9990577	2026-09-13 00:13	Initial checkpoint: SafeStep student + responder apps
1ab354d	2026-09-13 22:21	Add admin web dashboard, fix SOS reliability, and build real map + Walk With Me routing
ba399c2	2026-09-13 22:44	Add offline SOS fallback, fix map tiles for real, add location button
268a948	2026-09-16 13:10	Fix signup wizard race, per-student SOS ban, UTC timestamp bug, and zone-scoped alerts
6517f2e	2026-09-16 13:19	Add Safe Ride feature, rename Walk With Me to Safe Walks

1b93e23	2026-09-16 13:24	Fix group member names RLS gap and auto-scroll group messages
e08c6d0	2026-09-16 13:28	Expand privacy notice with full tiered-visibility content, add subtle NMU theme touch
a64b197	2026-09-16 13:41	Add theme toggle, SOS accessibility/haptics, and a visible support button
7ea52c6	2026-09-16 13:50	Add skeleton loaders, pull-to-refresh, better empty states, and undo on destructive actions
98e85bf	2026-09-16 13:56	Add biometric app-lock (Face/Touch ID / PIN fallback)
6271ab6	2026-09-16 17:11	Rebrand app to SafeStep: custom icon, branded auth screens, biometric lock escape hatch
1c3da2e	2026-09-16 17:11	Add home-screen SOS widget, fix SOS hold-button tap-vs-hold race

Two changes in this log are **not** represented in the commit history because they were live data operations rather than code changes: the student/staff login-email migration (100 real Supabase Auth accounts) and the Safety Resources content expansion (3 new + 2 updated guides, all seeded directly via the Supabase REST API under the service role).